HW 11

Monday, 27 March 2023 8:36 PM

Question 1

CSTR ; $B + \sigma S \rightarrow 2B + \dots$; K_s = michaelis constant

(B: Biomass; S: mass of substrate)

μ = max. specific growth

(i) Steady states? Desirable steady states? Inequality involving the operating parameter, θ and s_f?

Assuming perfectly-mixed : CSTR design equation $\frac{d(V c_j)}{dt} = q_{in} c_{jf} - q_{out} c_j + V \sum_{i=1}^{m} \alpha_{ij} r_i$

Constant V → $V \frac{dc_j}{dt} = q_{in} c_{jf} - q_{out} c_j + V r_B$

Mass equation (for B):

$$V \frac{db}{dt} = -q_{out} b + V r_B \quad (N_o\ b \text{ coming})$$

$$V \frac{db}{dt} = -q_{out} b + V\left(\frac{\mu s b}{K_s + s}\right)$$

$b = \frac{u s_f}{\sigma}$; $s = v \cdot s_f$; $t = \frac{\tau}{\mu}$; $K_s = \kappa \cdot s_f$

$$\Rightarrow V \frac{d\left(\frac{u s_f}{\sigma}\right)}{d\left(\frac{\tau}{\mu}\right)} = -q_{out}\left(\frac{u s_f}{\sigma}\right) + V\left(\frac{\mu \cdot v s_f \cdot \frac{u s_f}{\sigma}}{\kappa s_f + v s_f}\right)$$

$$s_f \frac{V\mu}{\sigma} \cdot \frac{du}{d\tau} = -q_{out} \cdot \frac{u s_f}{\sigma} + \frac{V}{\sigma}\left(\frac{\mu v \cdot u}{\kappa + v}\right) s_f$$

$$(V\mu) \cdot \frac{du}{d\tau} = -q_{out} \cdot u + V\left(\frac{\mu v \cdot u}{\kappa + v}\right)$$

$$V\mu \frac{du}{d\tau} = -\frac{V}{\theta} \cdot u + V\left(\frac{\mu v \cdot u}{\kappa + v}\right)$$

$$\Rightarrow \boxed{\frac{du}{d\tau} = -\frac{1}{(\mu\theta)} u + \left(\frac{v \cdot u}{\kappa + v}\right)}$$

$q = \frac{V}{\theta}$ → Total volume; → average residence time

Mass equation for C : $V \frac{ds}{dt} = q_{in} s_f - q_{out} s + V r_s$; $r_B = \frac{-r_s}{\sigma}$

$$V \frac{d(v \cdot s_f)}{d\left(\frac{\tau}{\mu}\right)} = q_{in} s_f - q_{out}(v \cdot s_f) - \frac{V}{\sigma}\left(\frac{\mu v \cdot u}{\kappa + v}\right) s_f \cdot \sigma$$

$$\mu V s_f \frac{dv}{d\tau} = s_f\left(q_{in} - q_{out} v\right) - V\left(\frac{\mu v \cdot u}{\kappa + v}\right) s_f$$

For CSTR: $q_{in} = q_{out} = \frac{V}{\theta}$

$$\therefore \mu V \frac{dv}{d\tau} = \frac{V}{\theta}(1 - v) - V\left(\frac{\mu v \cdot u}{\kappa + v}\right)$$

$$\boxed{\frac{dv}{d\tau} = \frac{1}{(\mu\theta)}(1 - v) - \left(\frac{v \cdot u}{\kappa + v}\right)}$$

$q = \frac{V}{\theta}$ → mean

$(\kappa + v) \sim v$.

@ SS $\frac{dv}{d\tau} = \frac{du}{d\tau} = 0 = F(u, v)$

Condition I : $v = 1$, $u = 0$ → No biomass formation, just substrate feed going in and out unreacted

↳ Undesirable

Condition II :

$$\frac{du}{d\tau} = -\frac{u}{\mu\theta} + \frac{v \cdot u}{\kappa + v} = 0$$

Solving for v : $\frac{v}{\kappa + v} = \frac{1}{\mu\theta}$

$$v(\mu\theta) = \kappa + v$$

Energy balance : $\frac{d\eta}{d\tau} = -\eta - \beta[\eta - \eta_c] + B Da\, R(\xi, \eta)$

Assuming adiabatic reactor : $\beta = 0$

$$\frac{d\eta}{d\tau} = -\eta + B Da\, R\left(\frac{\eta}{B}, \eta\right) \quad ; \quad B\xi = \eta$$

Specific steady state:

$$\eta = B Da\, R\left(\frac{\eta}{B}, \eta\right) \quad ; \quad R(\xi, \eta) =$$

CSTR Adiabatic energy balance: $T - T_f = \frac{(-\Delta H_R) X}{\sum \theta_i c_{i_f}}$;

$$\frac{r(c,T)}{r(c_i,T_j)} = \frac{r_a}{r_a^b} = \frac{\frac{\chi_a k}{K_a + c}}{\frac{\chi_a c_j k_j}{k_j^2 + c_j}} = \frac{sb}{(K_a + s)} \cdot \frac{(k_j^2 + c_j)}{c_j k_j}$$

$$X = \text{Conversion} = \frac{N_A^0 - N_j}{\alpha_j}$$

Solving for v: $\frac{v}{k+v} = \frac{1}{\mu\theta}$

$v(\mu\theta) = k+v$

$v(\mu\theta - 1) = k$

$v = \frac{k}{\mu\theta - 1}$ → Plug in v

$\frac{(1-v)}{\mu\theta} + \frac{v-u}{k+v} = 0$

Solving for u:

$\Rightarrow \frac{v-u}{k+v} = \frac{v-1}{\mu\theta}$

$u = \frac{(v-1)(k+v)}{v\mu\theta}$

$u = \frac{\left(\frac{k}{(\mu\theta-1)} - 1\right)\left(k + \frac{k}{(\mu\theta-1)}\right)}{v\mu\theta}$

$u = \frac{(k-(\mu\theta-1))(k(\mu\theta-1)+k)}{(\mu\theta-1)v\mu\theta}$

$u = \frac{(k-(1-\mu\theta))(1-\mu\theta)(-k)}{(\mu\theta-1)(v\mu\theta)}$

$u = \frac{(k+1) - \theta\mu}{(1-\theta\mu)}$

Condition II:

$(u^*, v^*) = \left[\frac{(k+1)-\theta\mu}{(1-\theta\mu)}, \frac{k}{(\mu\theta-1)}\right]$

↳ This state is desirable

Inequality for operation: $u > 0$

$\frac{(k+1)-\theta\mu}{(1-\theta\mu)} > 0$

(b) Evaluating stability. $F = \frac{du}{d\tau}$, $\frac{dv}{d\tau} = G$ vars (u, v)

Jacobian matrix:

$J = \begin{pmatrix} \frac{\partial F}{\partial u} & \frac{\partial F}{\partial v} \\ \frac{\partial G}{\partial u} & \frac{\partial G}{\partial v} \end{pmatrix}_{(u^*, v^*)}$

$\frac{\partial F}{\partial u} = -\frac{1}{\mu\theta} + \frac{v}{(k+v)}$ $\quad \frac{\partial G}{\partial u} = \frac{-v}{k+v}$

$\frac{\partial F}{\partial v} = \frac{(k+v)u - vu}{(k+v)^2}$ $\quad \frac{\partial G}{\partial v} = -\frac{1}{\mu\theta} - \left(\frac{(k+v)u - vu}{(k+v)^2}\right)$

Plugging in the values of steady state we can obtain determinant (Δ) and trace (τ):

For stability: $\tau^2 - 4\Delta < 0$, $\Delta \neq 0$ and $\tau^2 \neq 4\Delta$ & $\Delta < 0$

$\Delta = \det(J)$

$J = \begin{pmatrix} -\frac{1}{\mu\theta} + \frac{v}{(k+v)} & \frac{k}{(k+v)^2} \\ \frac{-v}{(k+v)} & -\frac{1}{\mu\theta} - \left(\frac{ku}{(k+v)^2}\right) \end{pmatrix}$

Condition 1: $v = 1$, $u = 0$

Condition 2: $v = \frac{k}{(\mu\theta-1)}$, $u = \frac{(k+1)-\theta\mu}{(1-\theta\mu)}$

$1 + u = \frac{1-\theta\mu+(k+1)-\theta\mu}{(1-\theta\mu)} = \frac{2+k-2\theta\mu}{(1-\theta\mu)}$

$\Rightarrow \frac{\frac{k}{(\mu\theta-1)}}{k + \frac{k}{(\mu\theta-1)}} \cdot \frac{1}{1} = \frac{\frac{k}{(\mu\theta-1)}}{\frac{k(\mu\theta-1)+k}{(\mu\theta-1)}} = \frac{k}{k(\mu\theta-1)+k}$

$= \frac{1}{(\mu\theta-1)+1}$

$= \frac{1}{\mu\theta}$

$J_1 = \begin{pmatrix} -\frac{1}{\mu\theta} + \frac{1}{(k+1)} & \frac{k}{(k+1)^2} \\ \frac{-1}{(k+1)} & -\frac{1}{\mu\theta} \end{pmatrix}$

$J_2 = \begin{pmatrix} 0 & \frac{1}{(\mu\theta)^2} \\ -\frac{1}{\mu\theta} & -\frac{1}{\mu\theta}\left(1 + \frac{k}{\mu\theta}\right) \end{pmatrix}$

$\Rightarrow \frac{k}{\mu\theta} = \frac{1}{\mu\theta}$

$v = \frac{\mu\theta'}{\sigma}$

$$J_2 = \begin{pmatrix} 0 & \frac{1}{(\mu\theta)^2} \\ -\frac{1}{\mu\theta} & -\frac{1}{\mu\theta}\left(1+\frac{\kappa}{(1+\kappa)}\right) \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \frac{1}{(\mu\theta)^2} \\ -\frac{1}{\mu\theta} & -\frac{1}{\mu\theta}\left(\frac{2+\kappa-2\theta\mu}{1-\theta\mu}\right) \end{pmatrix}$$

$$\det(J_1) = -\frac{1}{\mu\theta}\left(-\frac{1}{\mu\theta}+\frac{1}{(\kappa+1)}\right)+\frac{\kappa}{(\kappa+1)^3}$$

$$= \frac{1}{(\mu\theta)^2} - \frac{1}{\mu\theta(\kappa+1)} + \frac{\kappa}{(\kappa+1)^3}$$

$$\det(J_2) = \frac{1}{(\mu\theta)^2} > 0$$

$$\tau_{J_1} = -\frac{2}{\mu\theta}+\frac{1}{(\kappa+1)}$$

$$\tau_{J_2} = -\frac{1}{\mu\theta}\left(\frac{2+\kappa-2\theta\mu}{1-\mu\theta}\right)$$

(iii) Oscillations can exist when there are complex values for eigenvalues for the Jacobian.

$$= \frac{}{\mu\theta}$$

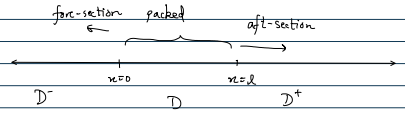
$$\Rightarrow \frac{\kappa}{\left(\kappa+\frac{\kappa}{(\mu\theta-1)}\right)\left(\kappa+\frac{\kappa}{(\mu\theta-1)}\right)} = \frac{1}{((\mu\theta-1)+1)((\mu\theta-1)+1)}$$

$$= \frac{1}{\mu\theta}$$

4

Question 2

PBR, axial dispersion model, No reaction; Tracer experiment



(i) From class notes:

(ii) each section has a general equation: $D^- \frac{\partial^2 c^-}{\partial x^2} - v_x \frac{\partial c^-}{\partial x} = \frac{\partial c^-}{\partial t}$; $c^- = \frac{u^- \cdot Q}{Al}$; $\tau = \frac{t \cdot Al}{q} = t$

$x = \xi \cdot l$; $v_x = \frac{Pe^- \cdot D^-}{l}$

$$\Rightarrow D^- \frac{\partial^2\left(\frac{u^- Q}{Al}\right)}{\partial(\xi l)^2} - \frac{Pe^- D^-}{l}\frac{\partial\left(\frac{u^- Q}{Al}\right)}{\partial(\xi l)} = \frac{\partial\left(\frac{u^- Q}{Al}\right)}{\partial\left(\frac{\tau \cdot Al}{q}\right)}$$

$$\frac{D^- Q}{Al}\cdot\frac{1}{l^2}\frac{\partial^2 u^-}{\partial\xi^2} - \frac{Pe^- D^-}{l}\frac{Q}{Al}\cdot\frac{1}{l}\frac{\partial u^-}{\partial\xi} = \frac{Q}{Al}\frac{q}{Al}\frac{\partial u^-}{\partial\tau}$$

$$\Rightarrow \frac{D^- Q}{Al^3}\frac{\partial^2 u^-}{\partial\xi^2} - \frac{Pe^- D^- Q}{Al^3}\frac{\partial u^-}{\partial\xi} = \frac{Qql}{Al^2 A}\frac{\partial u^-}{\partial\tau}$$

$$\Rightarrow D\frac{\partial^2 u^-}{\partial\xi^2} - Pe^- D^-\frac{\partial u^-}{\partial\xi} = \frac{lq}{A}\frac{\partial u^-}{\partial\tau}$$

$$\boxed{\frac{\partial^2 u^-}{\partial\xi^2} - Pe^-\frac{\partial u^-}{\partial\xi} = Pe^-\frac{\partial u^-}{\partial\tau}}$$

$Pe = \frac{lv}{D} = \frac{l\cdot q}{D\,A}$

Similarly

$$\boxed{\frac{\partial^2 u^+}{\partial\xi^2} - Pe^+\frac{\partial u^+}{\partial\xi} = Pe^+\frac{\partial u^+}{\partial\tau}}$$

$$\boxed{\frac{\partial^2 u}{\partial\xi^2} - Pe\frac{\partial u}{\partial\xi} = Pe\frac{\partial u}{\partial\tau}}$$

Boundary conditions: @ $t=0$ and $x=0$, $c^- = \frac{Q\delta(t)+c}{Al}$ — 2 BC & 1 IC

@ $\tau=0$, $\xi=0$, $u=u^- = \delta(\tau)$

$x=l$, $c=c^+$

@ $\xi=\infty$, $u=0$

$\Rightarrow$

@ $x = -\infty$, $c^- = 0$

@ $x = +\infty$, $c^+ = 0$

@ $\xi_L = +\infty$, $u^+ = 0$

(iii) Laplace transform of dimensionless residence time

$$\bar{f}(s) = \int_0^\infty f(t)\, e^{-st}\, dt$$

$$\mathcal{L}\left(\frac{\partial u}{\partial \tau}\right) = s\bar{u}(s) - u(0)$$

where $u^-(0) = \delta(\xi_L)$

$u^+(0) = 0$

$u(0) = \delta(\xi_L)$

$$\mathcal{L}\left(\frac{\partial u}{\partial \tau}\right) = \xi_L \frac{\partial \bar{u}(\xi_L, s)}{\partial \xi_L}.$$

⇒ Turning the PDE into ODE's:

$$\frac{d^2\bar{u}^-}{d\xi_L^2} - Pe\,\frac{d\bar{u}^-}{d\xi_L} - Pe\,s\,\bar{u}^- = 0$$

$$\frac{d^2\bar{u}}{d\xi_L^2} - Pe\,\frac{d\bar{u}}{d\xi_L} - Pe\,s\bar{u} = Pe\,\delta(\xi_L) \qquad (\text{for } + \text{ & normal } u)$$

↳ This can be solved numerically